

KEHEM-IT

PRESENTS

Shikkha ERP

একটি সম্পূর্ণ শিক্ষা ব্যবস্থাপনা প্ল্যাটফর্ম

Enterprise Education ERP Platform for Bangladesh

Core + Plugin Architecture · Multi-Institution · Multi-Curriculum · Bangla & English

SYSTEM ARCHITECTURE · FEATURE SPECIFICATION · MENU BLUEPRINT · DEVELOPMENT ROADMAP

Prepared for: BpyTutor / KEHEM-IT

Backend: Python (FastAPI) · PostgreSQL · VPS

Frontend: Vue 3 · TypeScript · SCSS/Tailwind · MastorsCDN · Font Awesome

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1. Executive Summary

Bangladesh's school-software market (Pathshala Soft/Zoom IT, Digital Pathshala, ClassTune, Fedena, ZoomIT and similar products) is dominated by single-tenant, feature-bundled systems built around attendance, exams, fees, and SMS notifications. They serve the market well but are architecturally monolithic: every feature ships together, customization is done per-client by the vendor, and government/board-specific logic (GPA calculation, MPO reporting, Madrasa streams) is hard-coded rather than configurable.

Shikkha ERP is proposed as a Core + Plugin Education ERP: a small, stable kernel (identity, institute/branch management, academic calendar, RBAC, notification bus, billing, and a plugin registry) around which every functional area — admissions, academics, exams, attendance, accounts, HR, library, transport, hostel, LMS, and even the public school website — is built as an independently installable, versioned plugin. This lets the product scale from a single Bangla-medium primary school to a multi-campus group running Bangla Medium, English Version, English Medium (Cambridge/Edexcel), Madrasa (Dakhil/Alim/Hifz), and College/HSC programs side by side, enabling or disabling modules per institution and per subscription tier.

Why Core + Plugin, not a monolith

- Institutions are heterogeneous: a Qawmi Madrasa doesn't need a GPA engine; an English Medium school doesn't need SSC/Dakhil board integration. Plugins let each institution run only what it needs.
- New features (e.g. a future NCTB curriculum change, a new board API, a new payment gateway) ship as a plugin update without touching or re-testing the whole system.
- A plugin marketplace becomes a second revenue stream — sell/rent modules (Transport, Hostel, LMS, Alumni) as add-ons per institution.
- Multi-tenant SaaS and single-tenant self-hosted deployments share the same core; only the plugin set and licensing differ.

Reference systems studied

Feature and workflow patterns were cross-checked against Pathshala Soft / Zoom IT (Pathshala.com), Digital Pathshala (Extreme Solutions, Chittagong), and general-purpose school ERPs (Fedena, ClassTune) — all of which converge on: admission & enrolment, attendance, exam/GPA result publication, fee collection via bKash/Nagad/SSLCommerz, SMS/parent communication, timetable, an inbuilt school website, and (increasingly) online classes.

2. Bangladesh Education System — Domain Context

The system must be modeled around how Bangladesh actually runs schools, not a generic international template. Key domain facts that drive the data model:

2.1 Institution types the ERP must support

Type	Notes
Government / MPO Primary & Secondary	Bangla-medium, NCTB curriculum, MPO (Monthly Pay Order) government subsidy reporting for teacher salaries
Bangla Medium (Non-MPO/Private)	NCTB curriculum, own fee structure
English Version	NCTB curriculum taught in English, follows Bangladesh boards (SSC/HSC)
English Medium	Cambridge (IGCSE/O & A Level) or Edexcel, own academic calendar & report format
Madrasa — Aliya	Ebtedayee → Dakhil → Alim → Fazil/Kamil, Bangladesh Madrasa Education Board (BMEB)
Madrasa — Qawmi / Hifz	Independent boards (Befaq), Hifz progress tracking (Para/Sipara completion), non-standard grading
College / Intermediate (HSC)	Science / Humanities / Business Studies groups, 4th subject rules, semester or yearly
Technical & Vocational	BTEB curriculum, trade/NTVQF competency tracking
Multi-campus Group	Shared brand, separate EIIN, separate finances, consolidated group reporting

2.2 Academic structure & assessment

- 11 Education Boards nationally: 9 General boards (Dhaka, Rajshahi, Chattogram, Barishal, Sylhet, Mymensingh, Jashore, Dinajpur, Cumilla) plus Madrasah (BMEB) and Technical (BTEB) boards.
- Public exams: SSC/Dakhil/SSC-Vocational at Class 10, HSC/Alim/HSC-Vocational at Class 12; internal school exams (1st/mid/pre-test/half-yearly/annual, model tests) feed continuous assessment.
- Grading: 8-point letter grading, A+ (GPA 5.00) down to F, with a per-subject pass mark and an official GPA formula (compulsory-subject average, 4th-subject bonus only if grade point > 2.00). This must be a configurable "Grading Scheme" plugin setting, not hard-coded, since internal exams, English Medium (percentage / Cambridge grades) and Hifz assessment all differ.
- EIIN (Educational Institution Identification Number) is the mandatory government identifier per institution and must be a first-class field used in every board/BANBEIS/EMIS export.

2.3 Payments & communication norms

- Fee collection must support bKash, Nagad, Rocket and SSLCommerz (card/net-banking aggregator) as first-class gateways, alongside bank challan and cash-at-counter reconciliation.
- SMS remains the primary parent-notification channel outside major cities (attendance, fee due, exam date, results); push notification and app inbox are secondary/complementary, not a replacement.
- Bilingual UI (বাংলা / English) is mandatory throughout — not just labels, but printable documents (admit card, certificate, tabulation sheet, ID card) in the language the institution certifies in.

3. Product Vision & Design Principles

3.1 Vision statement

Shikkha ERP is the operating system for a Bangladeshi educational institution: one login, one source of truth for students, staff, money and content, extensible by plugins, deployable as a single self-hosted VPS instance for one school or as a multi-tenant SaaS serving thousands.

3.2 Design principles

- Core stays thin. The core never contains domain features (no exam logic, no fee logic in core) — only identity, tenancy, RBAC, settings, events, and the plugin loader.
- Everything domain-specific is a plugin, including Website CMS, Exam & Result, Accounts, HR/Payroll, Library, Transport, Hostel, LMS, and Communication.
- Plugins communicate through the core's event bus and shared contracts (Student, Staff, AcademicYear, Class/Section IDs), never by reaching into each other's database tables directly.
- Configuration over customization: grading schemes, fee heads, admission workflows, ID card templates, and report layouts are configured per institution via the admin UI, not via code forks.
- Bangla-first, offline-tolerant: works over patchy connections common outside Dhaka; queues SMS/attendance sync when offline; PWA-capable frontend.
- API-first: every plugin exposes a versioned REST (and optionally GraphQL) API so the same core can power web, a future mobile app, and third-party integrations (BANBEIS/EMIS, banks).
- Security & data ownership: per-institution data isolation (schema-per-tenant or row-level tenant scoping), full audit trail, role-based field-level access (e.g. teachers can't see fee data).

4. System Architecture

4.1 High-level architecture

Three layers: Core Platform (kernel), Plugin Modules (features, independently deployable services or in-process apps), and Delivery Channels (Admin Web App, Public School Website, Parent/Student Portal, future Mobile App).

Layer	Responsibility	Examples
Core Kernel	Tenancy, Auth/RBAC, Institute/Branch, Academic Year, Plugin Registry, Event Bus, Settings, Audit Log, Billing/Licensing	auth-service, tenant-service, plugin-manager
Plugin Modules	All domain features, each with its own DB schema/tables, migrations, and permissions, registered against core via a manifest	admission, academics, exam-result, attendance, accounts, hr-payroll, library, transport, hostel, lms, communication, website-cms, inventory, alumni
Delivery Channels	UI surfaces consuming core + plugin APIs	Admin SPA (Vue3), Public Website (SSR/CMS), Parent/Student Portal, Mobile App (future), SMS/Email/Push gateways

4.2 Plugin contract (how a plugin integrates)

- plugin.json manifest — id, version, dependencies, required core version, exposed permissions, menu contributions, DB migration path.
- Lifecycle hooks — install(), migrate(), enable(tenant), disable(tenant), uninstall() — invoked by the Plugin Manager.
- Menu contribution — plugin declares its own menu/submenu items with required permission keys; Core merges them into the sidebar at runtime based on the logged-in user's role and the tenant's enabled plugins.
- Event subscriptions — e.g. Exam-Result plugin publishes result.published; Communication plugin subscribes and sends SMS; Accounts plugin publishes invoice.paid; Website-CMS subscribes to show a "results out" banner.
- Data contracts, not shared tables — plugins reference students/staff by core-issued UUIDs via API/event, never via direct foreign keys into another plugin's schema, so plugins can be enabled/disabled independently.

4.3 Deployment model (Python/PostgreSQL on VPS)

Component	Technology
API framework	Python 3.12, FastAPI (async), Pydantic v2 for schema validation
Task queue / background jobs	Celery + Redis (SMS dispatch, result PDF generation, report exports, scheduled MPO/board exports)
Database	PostgreSQL 16 — schema-per-tenant (self-hosted single school) or row-level tenant_id scoping (multi-tenant SaaS), Alembic migrations per plugin
Caching / sessions	Redis
Search	PostgreSQL full-text search initially; OpenSearch if scale requires

Component	Technology
File/media storage	S3-compatible object storage (e.g. self-hosted MinIO on VPS, or BDIX-friendly cloud) for documents, ID photos, certificates, CMS media
Web/reverse proxy	Nginx + Gunicorn/Uvicorn workers, Let's Encrypt TLS
Containerization	Docker Compose for single-VPS deployments; Docker + optional K3s for multi-tenant scale-out
CI/CD	GitHub Actions → build → test → deploy to VPS (or staging → production promotion)
Monitoring	Prometheus + Grafana, Sentry for error tracking, centralized logging (Loki)
Frontend	Vue 3 (Composition API) + TypeScript, Pinia for state, Vue Router, SCSS + Tailwind utility layer, MastorsCDN design tokens/components, Font Awesome icon set, Vite build
Public website rendering	Same Vue 3 app in SSR/SSG mode (Vite SSR or Nuxt-style) for SEO-friendly public school websites generated by the Website-CMS plugin
Realtime	WebSocket/SSE channel (attendance live status, online class notices, chat) via FastAPI + Redis pub/sub
Auth	JWT access + refresh tokens, OAuth2 password/social login for parents, optional SSO for staff, 2FA for admin roles

4.4 Multi-tenancy strategy

- Tier 1 — Self-hosted single institution: one VPS, schema-per-tenant not needed (single schema), simplest ops, sold as a one-time license or annual maintenance — mirrors how Pathshala/ZoomIT sell today.
- Tier 2 — Managed SaaS: one shared cluster, PostgreSQL schema-per-tenant for strong isolation, tenant provisioning automated by Core ("Institute Setup" wizard creates schema, seeds default plugins).
- Tier 3 — Multi-campus group: one tenant, multiple branches sharing a group-level dashboard but branch-level data partitioning (branch_id scoping), consolidated finance rollups.

5. The Core Module (Kernel)

The core ships in every deployment and cannot be disabled. It owns identity, structure, and plumbing that every plugin depends on.

5.1 Core capabilities

Capability	Description
Institute & Branch Management	Institution profile, EIIN, board affiliation, logo/letterhead, multi-branch/campus hierarchy
Academic Year & Calendar	Session/year setup, term/semester structure, holiday calendar, working-day calculation used by attendance & payroll
Class / Section / Group / Shift	Class (e.g. Class 6), Section (A/B), Group (Science/Commerce/Arts), Shift (Morning/Day), Version (Bangla/English)
User & Identity	Single login for Admin/Staff/Teacher/Student/Parent/Accountant, unified profile, password policy, 2FA
Role-Based Access Control (RBAC)	Predefined + custom roles, granular permission keys per menu/action, field-level masking (e.g. hide salary from non-HR roles)
Plugin Manager	Install/enable/disable/update plugins per tenant, dependency resolution, license/subscription check
Notification / Event Bus	Internal pub-sub for cross-plugin events; outward dispatch to SMS/Email/Push adapters
Settings Engine	Per-tenant configuration store (grading scheme, fee heads, ID card template, language, theme) consumed by plugins
Audit Log	Who changed what, when — required for financial and result-related actions
Billing / Licensing	Subscription plan, active plugin list, seat/branch limits, invoice-to-KEHEM-IT for SaaS tier
Document & Media Store	Central file storage service used by all plugins (photos, certificates, receipts, CMS assets)
Localization Engine	বাংলা/English string catalogue, Bangla numeral & date formatting, print-template localization

6. Feature Modules (Plugins)

Every module below is a self-contained plugin against the Core contract. Institutions license/enable only what they need.

6.1 Admission & Enrolment

Manages the full journey from inquiry to enrolled student.

- Online admission form builder (per class, per year), custom fields per institution type
- Online application fee payment (bKash/Nagad/SSLCommerz)
- Seat/quota management, waiting list, lottery/merit list generation
- Entrance test scheduling & mark entry, admission test admit card
- Document upload & verification (birth certificate, transfer certificate, photo)
- Auto-generation of student ID, roll number, section assignment on confirmation
- Sibling/quota discount rules linked to Accounts plugin

6.2 Student Information System (SIS)

The single source of truth for every enrolled student.

- 360° student profile: personal, guardian, address, previous institution, medical, documents
- Class/section/roll history across years, promotion & transfer workflow
- Guardian/parent linkage with portal access, multiple guardians support
- Transfer Certificate (TC) and Character Certificate generation
- Bulk import/export (CSV/Excel) for migration from legacy registers

6.3 Staff / HR & Payroll

Employee lifecycle for teaching and non-teaching staff.

- Staff profile, qualification, subject/class assignment, document store
- Leave management (casual/medical/earned), leave approval workflow
- Payroll with MPO government-scale support alongside private-scale pay, arrears, provident fund, festival bonus
- Teacher evaluation / appraisal module
- Substitute-teacher assignment on leave

6.4 Attendance

Daily and period-wise presence tracking for students and staff.

- Class-wise/section-wise daily attendance by teacher; period-wise attendance for secondary/college
- Biometric / RFID / QR device integration hooks; mobile self check-in for staff
- SMS/push auto-alert to parent on absence (via Communication plugin event)
- Minimum-attendance-to-sit-exam rule engine feeding Exam module
- Attendance analytics dashboard (trend, chronic absentee flag)

6.5 Academics & Curriculum

Subject, syllabus and timetable backbone.

- Curriculum/subject mapping per class/group (NCTB, Cambridge/Edexcel, Madrasa, BTEB trades)
- Syllabus & lesson-plan tracking per subject/teacher
- Timetable/routine builder with conflict detection (teacher/room clash)

- Homework/assignment posting, submission, and grading (also feeds LMS plugin)

6.6 Exam & Result Management

The most Bangladesh-specific plugin — configurable grading engine.

- Exam scheduling: 1st/mid-term, half-yearly, pre-test, test/model test, final/annual, public exam (SSC/HSC/Dakhil/Alim) mock series
- Configurable grading scheme per institution: 8-point GPA (A+–F) for NCTB streams, percentage/letter grades for Cambridge/Edexcel, Para/Sipara-based progress for Hifz
- Official GPA formula engine (compulsory subject average + conditional 4th-subject bonus), per-subject pass-mark enforcement
- Mark entry (teacher portal, spreadsheet-style bulk entry, mobile-friendly), moderation/approval before publish
- Auto tabulation sheet, mark sheet, and result card generation (Bangla + English), merit/rank list
- Result publish event → triggers SMS/portal notification to parents (Communication plugin) and optional public website results-lookup page (Website-CMS plugin)
- Re-scrutiny / "board challenge" style internal recheck request workflow

6.7 Fees & Accounts

Full financial control from fee structure to ledger.

- Configurable fee heads (tuition, admission, exam, transport, hostel, session, fine), class/group-wise fee structure, sibling/staff-child/merit discounts
- Invoice generation, partial payment, due reminders (SMS/portal)
- Payment gateways: bKash, Nagad, Rocket, SSLCommerz (cards/net banking), bank challan upload, cash-counter receipt with daily cash reconciliation
- General ledger, income/expense tracking, chart of accounts, MPO government-fund vs institution-fund segregation
- Financial reports: daily collection, outstanding dues, income-expenditure statement, balance sheet export for auditors

6.8 Library Management

Book and resource circulation.

- Catalogue with ISBN/barcode, multiple copies, category/shelf mapping
- Issue/return/renew workflow, fine calculation for late return
- Student/staff self-service search & reservation via portal

6.9 Transport Management

Fleet and route operations.

- Route & stoppage mapping, vehicle & driver assignment
- Student-to-route allocation with transport fee auto-linking to Accounts
- Optional GPS tracking integration and parent live-location view

6.10 Hostel Management

Residential facility operations.

- Room/seat allocation, hostel fee linkage to Accounts
- Attendance/roll-call, visitor log, leave-out register
- Mess/meal planning and billing (optional sub-module)

6.11 Inventory & Asset Management

Non-financial stock control.

- Store/stationery/lab equipment stock in-out, reorder alerts
- Asset register (furniture, computers, lab devices) with depreciation tag

6.12 Communication Hub

Every outbound message flows through one adapter layer.

- SMS gateway integration (local BD aggregators), bulk & triggered SMS templates
- Email and in-app/push notification, parent/student mobile inbox
- Notice board, circular publishing (targeted by class/section/role)
- Two-way parent-teacher messaging, PTM (parent-teacher meeting) scheduler

6.13 Online Classes & LMS

Blended and remote learning support.

- Live class scheduling with video integration (BBB/Jitsi/Zoom API), recording archive
- Digital content library (video/PDF/quiz) per subject/chapter
- Online quiz/assignment with auto-grading for objective questions
- Discussion/Q&A forum per class

6.14 Website CMS (Public School Website)

The institution's public presence, built and hosted from the same platform.

- Drag-and-drop page builder (home, about, admission, notices, gallery, faculty, results lookup, contact)
- Theme picker with per-institution branding (colors, logo, banner) built on MastorsCDN component/theme tokens
- Auto-syncing dynamic sections: latest notices, event calendar, exam results lookup (roll-based), staff directory — all pulled live from Core/plugin data, no duplicate content entry
- Blog/news, photo/video gallery, downloadable-forms library, multi-language (Bangla/English) page variants
- SEO fields (meta, sitemap, structured data) and custom-domain mapping per institution
- Separate mini-site per branch for multi-campus groups

6.15 Parent & Student Portal

Self-service channel, technically a thin app over other plugins' APIs.

- Attendance, homework, exam schedule & results, fee dues & online payment, notices, digital diary — one dashboard
- Multi-child view for parents with more than one enrolled child
- Downloadable admit card, ID card, certificates, receipts

6.16 ID Card & Certificate Studio

Document/print template designer.

- Drag-and-drop template designer for ID cards, admit cards, TC, character certificate, testimonials
- Bulk generation with QR/barcode for verification, print-ready PDF batch export

6.17 Discipline & Health Records

Non-academic student wellbeing tracking.

- Behavior/discipline incident log, counseling notes (restricted-access role)
- Health record: immunization, medical history, nurse visit log

6.18 Alumni Network

Post-graduation relationship management.

- Alumni directory opt-in, batch-wise grouping, event/reunion announcements
- Donation/fund tracking for alumni contributions

6.19 Reports & Analytics

Cross-plugin business intelligence.

- Institution dashboard: enrolment trend, attendance %, fee collection %, pass rate, staff ratio
- Configurable/exportable reports (PDF/Excel), scheduled email reports to management
- Group-level consolidated dashboard for multi-campus tenants

6.20 Government & Board Integration

Compliance-specific connectors, kept as isolated plugins so they can be updated independently as government systems change.

- BANBEIS / EMIS data export formats
- MPO teacher-salary reporting export
- Education Board result-verification/import connectors (SSC/HSC/Dakhil/Alim, per-board format)

7. Menu & Navigation Blueprint

The sidebar is assembled at runtime by the Core from each enabled plugin's menu contribution, filtered by the logged-in user's role permissions. Below is the target structure per primary role.

7.1 Super Admin (KEHEM-IT platform level — SaaS operators only)

Menu	Sub-menu items
Dashboard	Platform overview, active tenants, revenue, support tickets
Tenants	Institution list · Create institution · Suspend/Activate · Plan & billing
Plugin Marketplace	Installed plugins · Available plugins · Version releases · Pricing
Global Settings	SMS/Payment gateway credentials · Email templates · System-wide announcements
System Health	Server metrics · Job queue status · Error logs · Backups

7.2 Institute Admin

Menu	Sub-menu items
Dashboard	Enrolment, attendance %, fee collection, pass rate, notices widget
Institute Setup	Profile & EIIN · Branches/Campus · Academic Year · Class/Section/Group/Shift · Holidays · Grading Scheme
Admission	Applications · Admission Tests · Merit List · Seat/Quota · Admission Settings
Students	Student List · Add/Bulk Import · Promote/Transfer · ID Cards · TC & Certificates
Staff (HR)	Employee List · Attendance · Leave Requests · Payroll · Evaluation
Academics	Subjects & Curriculum · Timetable/Routine · Syllabus · Homework/Assignments
Attendance	Student Attendance · Staff Attendance · Attendance Reports
Exam & Result	Exam Schedule · Mark Entry · Mark Approval · Tabulation Sheet · Result Publish · Mark-sheet/Result Card Print
Accounts & Fees	Fee Structure · Invoices/Dues · Collect Payment · Expenses · Ledger/Reports
Library	Catalogue · Issue/Return · Fines
Transport	Routes & Vehicles · Student Allocation · Driver Assignment
Hostel	Rooms & Allocation · Roll Call · Mess/Meal Billing
Inventory	Stock In/Out · Asset Register
Communication	SMS/Notice Composer · Templates · Circulars · PTM Scheduler
Online Classes / LMS	Live Classes · Content Library · Quizzes · Assignments
Website	Pages · Theme & Branding · Notices/News · Gallery · Results Lookup · Domain Settings
Alumni	Directory · Events · Donations
Reports & Analytics	Academic Reports · Financial Reports · Attendance Reports · Custom/Scheduled Reports
Users & Roles	Staff/Parent/Student Accounts · Roles & Permissions · Audit Log

Menu	Sub-menu items
Plugins & Settings	Enabled Modules · Language & Theme · Notification Preferences · Subscription/Billing

7.3 Teacher

Menu	Sub-menu items
Dashboard	Today's classes, pending mark entry, homework due
My Classes	Class/Section list · Student roster
Attendance	Take Attendance (period/daily)
Academics	Syllabus Progress · Homework/Assignment · Lesson Plan
Exam	Mark Entry · Result View (own subjects)
Communication	Notices · Messages to Parents · PTM
Online Classes	Schedule/Join Live Class · Upload Content

7.4 Accountant

Menu	Sub-menu items
Dashboard	Today's collection, dues summary, expense summary
Fee Collection	Collect Payment · Invoice Search · Payment Gateway Reconciliation
Fee Structure	Fee Heads · Class-wise Structure · Discounts
Expenses	Record Expense · Vendor/Category
Ledger & Reports	General Ledger · Income-Expenditure · Outstanding Dues · Export
Payroll	Salary Processing · Payslips (if HR access granted)

7.5 Student / Parent Portal

Menu	Sub-menu items
Dashboard	Attendance %, upcoming exams, fee due, latest notices
Academics	Routine · Homework · Syllabus
Attendance	My/Child Attendance History
Exam & Result	Admit Card · Marks/Result · Mark Sheet Download
Fees	Due List · Pay Online · Payment History/Receipts
Library	My Issued Books · Search Catalogue
Transport / Hostel	My Route/Stoppage · My Room (if applicable)
Online Classes	Join Live Class · Content Library · Submit Assignment
Communication	Notices · Messages · Digital Diary
Profile	Personal Info · Documents · ID Card/Certificates Download

8. Core Data Model (High-Level Entities)

Illustrative, not exhaustive — every plugin adds its own tables that reference these core entities by ID.

Entity	Key Attributes
Institution	id, EIIN, name, board_affiliation, type (Bangla/English Version/English Medium/Madrassa/College), branches[]
Branch	id, institution_id, name, address
AcademicYear	id, institution_id, year_label, start_date, end_date, is_current
ClassLevel / Section / Group / Shift	id, name, ordering, curriculum_track
Student	id, student_uid, name (bn/en), dob, gender, guardian_ids[], current_class/section, admission_year, EIIN-linked roll history
Guardian	id, name, relation, phone, nid, portal_user_id
Staff	id, staff_uid, name, designation, subject_assignments[], employment_type (MPO/Private)
User / Role / Permission	id, role_key, permission_keys[], scope (institution/branch)
Subject / Curriculum	id, class_level_id, name, track (NCTB/Cambridge/Edexcel/Madrassa/BTEB)
Exam / MarkEntry / ResultCard	exam_id, subject_id, student_id, marks, grade, gpa, status
FeeHead / Invoice / Payment	fee_head_id, student_id, amount, due_date, gateway, status
Plugin / TenantPluginState	plugin_id, version, enabled_for_tenant, config_json
Event / AuditLog	event_type, payload, actor_id, timestamp, tenant_id

9. Non-Functional Requirements

Category	Requirement
Performance	Sub-2s page loads on 3G-equivalent connections; result publish/tabulation for 1,000+ students under 30s (async job)
Scalability	Single VPS supports one institution up to ~5,000 students comfortably; SaaS tier horizontally scales via containerized workers
Availability	99.5%+ uptime target for SaaS tier; automated daily encrypted backups, point-in-time recovery for self-hosted
Security	RBAC + field-level access, encrypted secrets, TLS everywhere, OWASP ASVS-aligned API hardening, rate limiting on auth/payment endpoints
Data Privacy	Guardian/student PII access logged and scoped; NID/financial data encrypted at rest
Localization	Full বাংলা + English UI and print templates; Bangla numerals/date formats configurable per institution
Offline tolerance	Attendance and mark-entry work offline-first in the browser (queued sync) for low-connectivity campuses
Accessibility	WCAG 2.1 AA target for parent/student portal and public website
Extensibility	New plugin can be installed without downtime; plugin API versioned to avoid breaking older mobile/portal clients
Compliance	BANBEIS/EMIS export formats kept current as an isolated, independently updatable plugin

10. Development Roadmap

Estimated for a small focused team (2–4 developers). Phases 2 onward can partially overlap once Core is stable; Website-CMS and Communication can start earlier in parallel if a second developer is available.

Phase	Duration	Deliverables
Phase 0 — Discovery & Foundations	3 weeks	Finalize domain model, grading-scheme spec, plugin contract & manifest schema, DB schema-per-tenant strategy, dev environment, CI/CD skeleton, MastorsCDN theme tokens applied to base UI kit
Phase 1 — Core Kernel	5 weeks	Auth/RBAC, Institute/Branch setup, Academic Year/Class/Section, Plugin Manager, Settings Engine, Event Bus, Audit Log, Admin shell UI (Vue3+TS)
Phase 2 — SIS, Admission & Attendance	6 weeks	Student Information System, Admission workflow, Attendance (student+staff), Guardian portal login
Phase 3 — Academics & Exam/Result	6 weeks	Subjects/Timetable/Homework, Exam scheduling, configurable Grading Engine (GPA/percentage/Hifz), mark entry, tabulation, result publish, printable mark sheets
Phase 4 — Fees & Accounts	5 weeks	Fee structure engine, invoicing, bKash/Nagad/SSLCommerz integration, ledger & financial reports
Phase 5 — Communication & Website CMS	5 weeks	SMS/Email/Push adapter layer, notice/circular system, Website-CMS page builder + theme engine, results-lookup public page
Phase 6 — HR/Payroll, Library, Transport, Hostel, Inventory	6 weeks	Remaining operational plugins, each independently toggleable
Phase 7 — LMS / Online Classes & ID Card Studio	4 weeks	Live class integration, content library, quiz engine, template designer for ID cards/certificates
Phase 8 — Reports, Analytics & Government/Board Connectors	4 weeks	Cross-plugin dashboards, BANBEIS/EMIS/MPO export plugins, board result-verification connectors
Phase 9 — Multi-tenant SaaS hardening	4 weeks	Schema-per-tenant provisioning automation, billing/subscription, plugin marketplace UI, load testing
Phase 10 — Pilot, QA & Launch	4 weeks	Pilot with 2–3 real institutions (mix of Bangla-medium + Madrasa + English Medium to validate plugin flexibility), bug-bash, security audit, go-live

Total sequential estimate: ~52 weeks (~12 months) for full scope with one team. Running Phases 3, 5 and 6 partially in parallel with a 2-track team can realistically bring a sellable MVP (Core + SIS + Attendance + Exam/Result + Fees + Website) to market in 4–5 months, with remaining plugins layered on afterward — matching how the reference products in this market actually shipped (core academic+fee first, transport/hostel/LMS later).

10.1 Suggested MVP scope (first sellable release)

- Core Kernel + Institute Setup + RBAC
- Student Information System + Admission
- Attendance (student)
- Exam & Result with configurable GPA engine
- Fees & Accounts with bKash/Nagad

- SMS Communication
- Basic Website CMS (institution can go live with a public site day one)

10.2 Team shape

Role	Focus
Backend Lead (Python/FastAPI/Postgres)	Core kernel, plugin contract, DB design
Backend Developer	Plugin implementation (Exam/Accounts/Attendance)
Frontend Lead (Vue3/TS)	Admin shell, design system integration (MastorsCDN/Tailwind), component library
Frontend Developer	Portal + Website-CMS builder UI
QA / DevOps (part-time)	CI/CD, testing, VPS deployment, security review

11. Key Risks & Mitigations

Risk	Mitigation
Grading/board rules change or vary per board	Isolate as a configurable Grading Scheme + Board Connector plugin, not hard-coded in core
Payment gateway API changes (bKash/Nagad/SSLCommerz)	Adapter pattern per gateway behind a common Payment interface
Plugin sprawl / dependency conflicts	Strict plugin manifest with declared core-version compatibility, automated compatibility checks in Plugin Manager
Data isolation failure in multi-tenant SaaS	Schema-per-tenant by default; automated tests asserting cross-tenant query isolation
Low-connectivity usage outside Dhaka	Offline-first attendance/mark-entry with sync queue; SMS as fallback channel
Feature creep vs. reference competitors	Lock MVP scope (Section 10.1) before adding transport/hostel/LMS depth

12. Appendix — Reference Comparison

System	Architecture	Notable strengths	Gap Shikkha ERP addresses
Pathshala Soft / Zoom IT	Monolithic, vendor-hosted	Strong local market trust, bKash/Nagad integration, inbuilt website per branch	No plugin/marketplace model; customization is vendor-manual, not self-serve config
Digital Pathshala (Extreme Solutions)	Monolithic, web-based	Multi-branch, integrated accounting	Same monolithic limitation; not extensible per institution type
Fedena / ClassTune (generic)	Configurable but not Bangladesh-specific by default	Broad feature set, higher-ed course/credit support	Requires heavy localization for GPA/board/MPO/bKash workflows
Shikkha ERP (proposed)	Core + Plugin, API-first	Bangladesh-specific core (GPA engine, EIIN, MPO, bKash/Nagad, Madrasa tracks) with a true plugin marketplace and self-serve configuration	—

This document is a planning blueprint. Before Phase 0 begins, validate the grading-scheme and fee-structure configuration schemas against 2–3 real target institutions (one Bangla-medium MPO school, one English Medium school, one Madrasa) to make sure the "configurable, not hard-coded" principle actually holds for their real workflows.